

Understanding HTML5

The Backbone of the Modern Web

1. Introduction to HTML

HTML, or **HyperText Markup Language**, is the standard markup language used to create web pages. It describes the structure of a webpage semantically and originally included cues for the appearance of the document.

Key Concept: HTML is not a programming language; it is a markup language. It tells the browser how to display content like text, images, and links by wrapping them in "tags".

The Basic Document Structure

Every HTML5 document follows a specific skeleton to be recognized correctly by web browsers:

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  <h1>This is a Heading</h1>
  <p>This is a paragraph.</p>
</body>
</html>
```

Core Components

- **<!DOCTYPE html>**: Defines that this document is an HTML5 document.
- **<html>**: The root element of an HTML page.
- **<head>**: Contains meta information about the page (title, character set, styles).
- **<body>**: Contains the visible page content.

2. Common HTML Elements

Elements are the building blocks of HTML. Most elements have an opening tag, some content, and a closing tag.

Text Formatting

Tag	Description
<h1> to <h6>	Defines HTML headings (h1 is the most important).
<p>	Defines a paragraph.
	Defines important text (usually rendered as bold).
	Defines emphasized text (usually rendered as italic).

Links and Images

The web is built on "Hyperlinks". To link to another page, we use the anchor tag:

```
<a href="https://www.google.com">Visit Google</a>
```

To display an image, we use the image tag, which is "self-closing" (it doesn't need a tag):

```

```

Lists

HTML supports both ordered (numbered) and unordered (bulleted) lists:

```
<ul>
  <li>First item</li>
  <li>Second item</li>
</ul>
```

3. HTML5 Semantic Elements

Modern HTML (HTML5) introduced semantic elements. These elements clearly describe their meaning to both the browser and the developer.

Why use Semantic Elements? They improve Accessibility (for screen readers) and SEO (Search Engine Optimization), helping search engines understand the structure of your content.

Standard Layout Elements

- **<header>**: Defines a header for a document or a section.
- **<nav>**: Defines a set of navigation links.
- **<main>**: Specifies the main content of a document.
- **<section>**: Defines a section in a document.
- **<article>**: Defines an independent, self-contained content block.
- **<aside>**: Defines content aside from the page content (like a sidebar).
- **<footer>**: Defines a footer for a document or a section.

The Future of HTML

HTML continues to evolve. With the integration of the Web Components API, developers can now create custom, reusable HTML tags. Furthermore, HTML works hand-in-hand with CSS (for styling) and JavaScript (for interactivity) to create the complex web applications we use today.